

TP KAFKA — Compte-rendu

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Partie 2 — Installation de Kafka et premier démarrage

Exercice 2.A — Premier démarrage

Captures

Démarrez le broker en avant-plan.

```
[2026-05-02 20:09:12,965] INFO [BrokerLifecycleManager id=1] Successfully registered broker 1 with broker epoch 6 (kafka.server.BrokerLifecycleManager)
[2026-05-02 20:09:12,966] INFO Loading logs from log dirs ArrayBuffer(C:\kafka-data\logs) (kafka.log.LogManager)
[2026-05-02 20:09:12,974] INFO No logs found to be loaded in C:\kafka-data\logs (kafka.log.LogManager)
[2026-05-02 20:09:12,981] INFO Loaded 0 logs in 14ms (kafka.log.LogManager)
[2026-05-02 20:09:12,982] INFO Starting log cleanup with a period of 300000 ms. (kafka.log.LogManager)
[2026-05-02 20:09:12,984] INFO Starting log flusher with a default period of 9223372036854775807 ms. (kafka.log.LogManager)
[2026-05-02 20:09:13,035] INFO [kafka-log-cleaner-thread-0]: Starting (kafka.log.LogCleaner$CleanerThread)
[2026-05-02 20:09:13,040] INFO [GroupCoordinator 1]: Starting up. (kafka.coordinator.group.GroupCoordinator)
[2026-05-02 20:09:13,040] INFO [AddPartitionsToTxnSenderThread-1]: Starting (kafka.server.AddPartitionsToTxnManager)
[2026-05-02 20:09:13,040] INFO [LogDirFailureHandler]: Starting (kafka.server.ReplicaManager$LogDirFailureHandler)
[2026-05-02 20:09:13,043] INFO [GroupCoordinator 1]: Startup complete. (kafka.coordinator.group.GroupCoordinator)
[2026-05-02 20:09:13,045] INFO [TransactionCoordinator id=1] Starting up. (kafka.coordinator.transaction.TransactionCoordinator)
[2026-05-02 20:09:13,048] INFO [TxnMarkerSenderThread-1]: Starting (kafka.coordinator.transaction.TransactionMarkerChannelManager)
[2026-05-02 20:09:13,048] INFO [TransactionCoordinator id=1] Startup complete. (kafka.coordinator.transaction.TransactionCoordinator)
[2026-05-02 20:09:13,059] INFO [MetadataLoader id=1] InitializeNewPublishers: initializing BrokerRegistrationTracker(id=1) with a snapshot at offset 5 (org.apache.kafka.image.loader.MetadataLoader)
[2026-05-02 20:09:13,071] INFO [BrokerLifecycleManager id=1] The broker has caught up. Transitioning from STARTING to RECOVERY. (kafka.server.BrokerLifecycleManager)
[2026-05-02 20:09:13,072] INFO [BrokerServer id=1] Finished waiting for the controller to acknowledge that we are caught up (kafka.server.BrokerServer)
[2026-05-02 20:09:13,074] INFO [BrokerServer id=1] Waiting for the initial broker metadata update to be published (kafka.server.BrokerServer)
[2026-05-02 20:09:13,075] INFO [BrokerServer id=1] Finished waiting for the initial broker metadata update to be published (kafka.server.BrokerServer)
[2026-05-02 20:09:13,076] INFO KafkaConfig values:
    advertised.listeners = PLAINTEXT://localhost:9092,CONTROLLER://localhost:9093
```

liste des versions API supportées.

```
PS C:\WINDOWS\system32> $env:Path = "C:\kafka\bin\windows;$env:Path"
>> kafka-broker-api-versions.bat --bootstrap-server localhost:9092
localhost:9092 (id: 1 rack: null) -> (
  Produce(0): 0 to 11 [usable: 11],
  Fetch(1): 0 to 17 [usable: 17],
  ListOffsets(2): 0 to 9 [usable: 9],
  Metadata(3): 0 to 12 [usable: 12],
  LeaderAndIsr(4): UNSUPPORTED,
  StopReplica(5): UNSUPPORTED,
  UpdateMetadata(6): UNSUPPORTED,
  ControlledShutdown(7): UNSUPPORTED,
  OffsetCommit(8): 0 to 9 [usable: 9],
  OffsetFetch(9): 0 to 9 [usable: 9],
  FindCoordinator(10): 0 to 6 [usable: 6],
  JoinGroup(11): 0 to 9 [usable: 9],
  Heartbeat(12): 0 to 4 [usable: 4],
  LeaveGroup(13): 0 to 5 [usable: 5],
  SyncGroup(14): 0 to 5 [usable: 5],
  DescribeGroups(15): 0 to 5 [usable: 5],
  ListGroups(16): 0 to 5 [usable: 5],
  SaslHandshake(17): 0 to 1 [usable: 1],
  ApiVersions(18): 0 to 4 [usable: 4],
  CreateTopics(19): 0 to 7 [usable: 7],
  DeleteTopics(20): 0 to 6 [usable: 6],
  DeleteRecords(21): 0 to 2 [usable: 2],
  InitProducerId(22): 0 to 5 [usable: 5],
  OffsetForLeaderEpoch(23): 0 to 4 [usable: 4],
  AddPartitionsToTxn(24): 0 to 5 [usable: 5],
  AddOffsetsToTxn(25): 0 to 4 [usable: 4],
  EndTxn(26): 0 to 4 [usable: 4],
  WriteTxnMarkers(27): 0 to 1 [usable: 1],
  TxnOffsetCommit(28): 0 to 4 [usable: 4],
  DescribeAcls(29): 0 to 3 [usable: 3],
  CreateAcls(30): 0 to 3 [usable: 3],
  DeleteAcls(31): 0 to 3 [usable: 3],
  DescribeConfigs(32): 0 to 4 [usable: 4],
  AlterConfigs(33): 0 to 2 [usable: 2],
  AlterReplicaLogDirs(34): 0 to 2 [usable: 2],
```

cluster-id généré.

```
PS C:\WINDOWS\system32> kafka-storage.bat random-uuid
Mx20myAcRueNCzcjmPPyqA
```

Partie 3 — Manipulation en ligne de commande

Exercice 3.A — Création et exploration

captures

```

PS C:\WINDOWS\system32> kafka-topics.bat --bootstrap-server localhost:9092 --list
ventes
PS C:\WINDOWS\system32> kafka-topics.bat --bootstrap-server localhost:9092 --describe --topic ventes
Topic: ventes    TopicId: 6FSfvaF5TtC6Yg8Muy0W7Q PartitionCount: 3      ReplicationFactor: 1    Configs: segment.bytes=1073741824
    Topic: ventes    Partition: 0      Leader: 1      Replicas: 1      Isr: 1 Elr:      LastKnownElr:
    Topic: ventes    Partition: 1      Leader: 1      Replicas: 1      Isr: 1 Elr:      LastKnownElr:
    Topic: ventes    Partition: 2      Leader: 1      Replicas: 1      Isr: 1 Elr:      LastKnownElr:
PS C:\WINDOWS\system32> kafka-topics.bat --bootstrap-server localhost:9092 --create --topic logs-application --partition
s 4 --replication-factor 1
Created topic logs-application.
PS C:\WINDOWS\system32> kafka-topics.bat --bootstrap-server localhost:9092 --describe --topic logs-application
Topic: logs-application TopicId: Ti6dj1aYQKmA3TA PartitionCount: 4      ReplicationFactor: 1    Configs: segment
.bytes=1073741824
    Topic: logs-application Partition: 0      Leader: 1      Replicas: 1      Isr: 1 Elr:      LastKnownElr:
    Topic: logs-application Partition: 1      Leader: 1      Replicas: 1      Isr: 1 Elr:      LastKnownElr:
    Topic: logs-application Partition: 2      Leader: 1      Replicas: 1      Isr: 1 Elr:      LastKnownElr:
    Topic: logs-application Partition: 3      Leader: 1      Replicas: 1      Isr: 1 Elr:      LastKnownElr:
PS C:\WINDOWS\system32> kafka-topics.bat --bootstrap-server localhost:9092 --describe --topic logs-application
Topic: logs-application TopicId: Ti6dj1aYQKmA3TA PartitionCount: 4      ReplicationFactor: 1    Configs: segment
.bytes=1073741824
    Topic: logs-application Partition: 0      Leader: 1      Replicas: 1      Isr: 1 Elr:      LastKnownElr:
    Topic: logs-application Partition: 1      Leader: 1      Replicas: 1      Isr: 1 Elr:      LastKnownElr:
    Topic: logs-application Partition: 2      Leader: 1      Replicas: 1      Isr: 1 Elr:      LastKnownElr:
    Topic: logs-application Partition: 3      Leader: 1      Replicas: 1      Isr: 1 Elr:      LastKnownElr:
PS C:\WINDOWS\system32> kafka-console-producer.bat --bootstrap-server localhost:9092 --topic logs-application
>msg-1
>msg-2
>msg-3
>msg-4
>msg-5
>msg-6
>msg-7
>msg-8
>msg-9
>msg-10
>Terminer le programme de commandes (O/N) ? o
PS C:\WINDOWS\system32> kafka-console-consumer.bat --bootstrap-server localhost:9092 --topic logs-application --group an
alyseur-1 --from-beginning --property "print.partition=true"
Partition:0      msg-1
Partition:0      msg-2
Partition:0      msg-3
Partition:0      msg-4
Partition:0      msg-5
Partition:0      msg-6
Partition:0      msg-7
Partition:0      msg-8
Partition:0      msg-9
Partition:0      msg-10
Processed a total of 10 messages

```

Les messages sont-ils répartis équitablement ?

Non, ils sont tous dans la partition 0. Sans clé, Kafka utilise le sticky partitioner qui regroupe les messages en batch vers la même partition.

Exercice 3.B — Effet de la clé sur le partitionnement

```

PS C:\WINDOWS\system32> kafka-console-producer.bat --bootstrap-server localhost:9092 --topic logs-application --property "parse.key=true" --property "key.separator=:"
>serveur-1:log-A1
>serveur-1:log-A2
>serveur-1:log-A3
serveur-1:log-A4
serveur-2:log-B1
serveur-2:log-B2
serveur-2:log-B3
serveur-2:log-B4
serveur-3:log-C1
serveur-3:log-C2
serveur-3:log-C3

```

```
PS C:\WINDOWS\system32> kafka-console-consumer.bat --bootstrap-server localhost:9092 --topic logs-application --from-beginning --property print.key=true --property print.partition=true --property print.offset=true
Partition:1 Offset:0 serveur-1 log-A1
Partition:1 Offset:1 serveur-1 log-A2
Partition:1 Offset:2 serveur-1 log-A3
Partition:1 Offset:3 serveur-1 log-A4
Partition:0 Offset:0 null msg-1
Partition:0 Offset:1 null msg-2
Partition:0 Offset:2 null msg-3
Partition:0 Offset:3 null msg-4
Partition:0 Offset:4 null msg-5
Partition:0 Offset:5 null msg-6
Partition:0 Offset:6 null msg-7
Partition:0 Offset:7 null msg-8
Partition:0 Offset:8 null msg-9
Partition:0 Offset:9 null msg-10
Partition:3 Offset:0 serveur-2 log-B1
Partition:3 Offset:1 serveur-2 log-B2
Partition:3 Offset:2 serveur-2 log-B3
Partition:3 Offset:3 serveur-2 log-B4
Partition:2 Offset:0 serveur-3 log-C1
Partition:2 Offset:1 serveur-3 log-C2
Partition:2 Offset:2 serveur-3 log-C3
```

Question 1 : Tous les messages d'une même clé arrivent-ils dans la même partition ?

Oui

Clé	Partition
`serveur-1`	1
`serveur-2`	3
`serveur-3`	2

Question 2 : Que se passerait-il si on augmentait à 8 partitions ?

Le mapping change

Clé	Avec 4 partitions	Avec 8 partitions
`serveur-1`	1	`hash("serveur-1") % 8` = probablement **5**
`serveur-2`	3	`hash("serveur-2") % 8` = probablement **7**
`serveur-3`	2	`hash("serveur-3") % 8` = probablement **6**

Exercice 3.C — Groupes de consommateurs et parallélisme

```
PS C:\WINDOWS\system32> kafka-console-consumer.bat --bootstrap-server localhost:9092 --topic logs-application --group groupe-A --from-beginning
log-A1
log-A2
log-A3
log-A4
msg-1
msg-2
msg-3
msg-4
msg-5
msg-6
msg-7
msg-8
msg-9
msg-10
log-B1
log-B2
log-B3
log-B4
log-C1
log-C2
...
```

```
PS C:\WINDOWS\system32> kafka-consumer-groups.bat --bootstrap-server localhost:9092 --describe --group groupe-A
```

GROUP	TOPIC	PARTITION	CURRENT-OFFSET	LOG-END-OFFSET	LAG	CONSUMER-ID	HOST	CLIENT-ID
groupe-A	logs-application	1	4	4	0	console-consumer-598d1226-01f8-42fe-ac77-6911d7f8076a	/127.0.0.1	console-consumer
groupe-A	logs-application	0	10	10	0	console-consumer-598d1226-01f8-42fe-ac77-6911d7f8076a	/127.0.0.1	console-consumer
groupe-A	logs-application	3	4	4	0	console-consumer-598d1226-01f8-42fe-ac77-6911d7f8076a	/127.0.0.1	console-consumer
groupe-A	logs-application	2	4	4	0	console-consumer-598d1226-01f8-42fe-ac77-6911d7f8076a	/127.0.0.1	console-consumer

```
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kafka-consumer-groups.bat --bootstrap-server localhost:9092 --describe --group groupe-A
```

GROUP	TOPIC	PARTITION	CURRENT-OFFSET	LOG-END-OFFSET	LAG	CONSUMER-ID
	HOST					CLIENT-ID
groupe-A	logs-application	1	4	4	0	console-consumer-9c51a681b2-44fe-a433-601d887d1bd0 /127.0.0.1
groupe-A	logs-application	0	10	10	0	console-consumer-9c51a681b2-44fe-a433-601d887d1bd0 /127.0.0.1
groupe-A	logs-application	3	4	4	0	console-consumer-9c51a681b2-44fe-a433-601d887d1bd0 /127.0.0.1
groupe-A	logs-application	2	4	4	0	console-consumer-9c51a681b2-44fe-a433-601d887d1bd0 /127.0.0.1

```
PS C:\WINDOWS\system32> kafka-console-producer.bat --bootstrap-server localhost:9092 --topic logs-application
```

```
>test-1
>test-2
>test-3
>test-4
>test-5
>test-6
>test-7
>test-8
>Terminer le programme de commandes (O/N) ? o
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> C:\kafka\bin\windows\kafka-consumer-groups.bat --bootstrap-server localhost:9092 --describe --group groupe-A
```

GROUP	TOPIC	PARTITION	CURRENT-OFFSET	LOG-END-OFFSET	LAG	CONSUMER-ID
	HOST					CLIENT-ID
groupe-A	logs-application	3	4	4	0	console-consumer-b0d9da12-ec07-406f-8098-455b0ddf7fda /127.0.0.1
groupe-A	logs-application	2	12	12	0	console-consumer-b0d9da12-ec07-406f-8098-455b0ddf7fda /127.0.0.1
groupe-A	logs-application	1	4	4	0	console-consumer-70d02c0f-cd1b-4604-9915-0a23d20858ca /127.0.0.1
groupe-A	logs-application	0	110	110	0	console-consumer-70d02c0f-cd1b-4604-9915-0a23d20858ca /127.0.0.1

```
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> C:\kafka\bin\windows\kafka-consumer-groups.bat --bootstrap-server localhost:9092 --describe --group groupe-A

GROUP          TOPIC          HOST          PARTITION  CURRENT-OFFSET  LOG-END-OFFSET  LAG          CONSUMER-ID
groupe-A      logs-application  1             4           4               4              0           console-consumer-70d02c0f-cd
fb-4604-9915-0a23d20858ca /127.0.0.1
groupe-A      logs-application  0             110          110             0              0           console-consumer-70d02c0f-cd
fb-4604-9915-0a23d20858ca /127.0.0.1
groupe-A      logs-application  2             12           12              0              0           console-consumer-b0d9da12-ec
07-406f-8098-455b0ddf7fda /127.0.0.1
groupe-A      logs-application  3             4            4              0              0           console-consumer-c74fb27a-58
3e-47f8-b7cc-2f8e15a34249 /127.0.0.1
PS C:\WINDOWS\system32>
```

Question 1 :

Configuration	Répartition
2 consumers	2 partitions chacun
3 consumers	2 + 1 + 1 partitions

Question 2 — 5ème consumer :

Le 5ème consumer rejoint le groupe mais reçoit 0 partition — il reste complètement inactif. Il n'y a que 4 partitions, donc le 5ème ne sert à rien.

Partie 4 — Programmation Producer / Consumer en Java


```
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-lang3/3.19.0/commons-lang3-3.19.0.jar (709 kB at 24 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/com/ibm/icu/icu4j/77.1/icu4j-77.1.jar (15 MB at 4 MB at 194 kB/s)
[INFO] Generating project in Batch mode
Downloading from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml
Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (18 MB at 28 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/archetypes/maven-archetype-quickstart-1.4.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/archetypes/maven-archetype-quickstart-1.4.jar (7.1 kB at 39 kB/s)
[INFO] -----
[INFO] Using following parameters for creating project from Archetype: maven-archetype-quickstart:1.4
[INFO] -----
[INFO] Parameter: groupId, Value: tn.utm.kafka
[INFO] Parameter: artifactId, Value: tp-kafka-java
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] Parameter: package, Value: tn.utm.kafka
[INFO] Parameter: packageInPathFormat, Value: tn/utm/kafka
[INFO] Parameter: package, Value: tn.utm.kafka
[INFO] Parameter: groupId, Value: tn.utm.kafka
[INFO] Parameter: artifactId, Value: tp-kafka-java
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] Project created from Archetype in dir: C:\tp-kafka-java
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 50.044 s
[INFO] Finished at: 2026-05-02T21:11:47+01:00
[INFO] -----
PS C:\>
```

```
pom - Bloc-notes
Fichier Edition Format Affichage Aide
<?xml version="1.0" encoding="UTF-8"?>

<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion>

  <groupId>tn.utm.kafka</groupId>
  <artifactId>tp-kafka-java</artifactId>
  <version>1.0-SNAPSHOT</version>

  <name>tp-kafka-java</name>
  <!-- FIXME change it to the project's website -->
  <url>http://www.example.com</url>

  <properties>
    <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
    <maven.compiler.source>1.7</maven.compiler.source>
    <maven.compiler.target>1.7</maven.compiler.target>
  </properties>

  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>4.11</version>
      <scope>test</scope>
    </dependency>
  </dependencies>
</project>
```

Ln 1, Col 1 100% Windows (CRLF) UTF-8

```

PS C:\WINDOWS\system32> cd C:\tp-kafka-java
>> mvn exec:java "-Dexec.mainClass=tn.utm.kafka.SimpleProducer"
[INFO] Scanning for projects...
[INFO]
[INFO] -----< tn.utm.kafka:tp-kafka-java >-----
[INFO] Building tp-kafka-java 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- exec:3.1.0:java (default-cli) @ tp-kafka-java ---
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
? Envoyé ? partition=0, offset=0, key=client-1
? Envoyé ? partition=0, offset=1, key=client-1
? Envoyé ? partition=0, offset=2, key=client-1
? Envoyé ? partition=0, offset=3, key=client-1
? Envoyé ? partition=2, offset=0, key=client-0
? Envoyé ? partition=2, offset=1, key=client-0
? Envoyé ? partition=2, offset=2, key=client-0
? Envoyé ? partition=1, offset=0, key=client-2
? Envoyé ? partition=1, offset=1, key=client-2
? Envoyé ? partition=1, offset=2, key=client-2
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.194 s
[INFO] Finished at: 2026-05-02T21:21:34+01:00
[INFO] -----
PS C:\tp-kafka-java>

```

```

[INFO]
[INFO] --- exec:3.1.0:java (default-cli) @ tp-kafka-java ---
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
? En attente de messages... (Ctrl+C pour arrêter)
? partition=0, offset=0, key=client-1, value=Achat numéro 1 (67 DT)
? partition=0, offset=1, key=client-1, value=Achat numéro 4 (118 DT)
? partition=0, offset=2, key=client-1, value=Achat numéro 7 (169 DT)
? partition=0, offset=3, key=client-1, value=Achat numéro 10 (220 DT)
? partition=2, offset=0, key=client-0, value=Achat numéro 3 (101 DT)
? partition=2, offset=1, key=client-0, value=Achat numéro 6 (152 DT)
? partition=2, offset=2, key=client-0, value=Achat numéro 9 (203 DT)
? partition=1, offset=0, key=client-2, value=Achat numéro 2 (84 DT)
? partition=1, offset=1, key=client-2, value=Achat numéro 5 (135 DT)
? partition=1, offset=2, key=client-2, value=Achat numéro 8 (186 DT)

```

Question : Si vous lancez DEUX instances du consumer (dans le même groupe-java-1)

Le topic ventes a 3 partitions actives (0, 1, 2), donc :

Consumer	Partitions assignées
Consumer 1	Partition 0 + Partition 1
Consumer 2	Partition 2

Exercice 4.B — Idempotence et fiabilité

```
cally
[INFO] --- resources:3.4.0:testResources (default-testResources) @ tp-kafka-java ---
[INFO] skip non existing resourceDirectory C:\tp-kafka-java\src\test\resources
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ tp-kafka-java ---
[INFO] Recompiling the module because of changed dependency.
[INFO] --- surefire:3.5.4:test (default-test) @ tp-kafka-java ---
[INFO] --- jar:3.5.0:jar (default-jar) @ tp-kafka-java ---
[INFO] Building jar: C:\tp-kafka-java\target\tp-kafka-java-1.0-SNAPSHOT.jar
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 2.411 s
[INFO] Finished at: 2026-05-02T21:25:42+01:00
[INFO] -----
PS C:\tp-kafka-java>
```

```
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
? Envoyé ? partition=0, offset=4, key=client-1
? Envoyé ? partition=0, offset=5, key=client-1
? Envoyé ? partition=0, offset=6, key=client-1
? Envoyé ? partition=0, offset=7, key=client-1
? Envoyé ? partition=2, offset=3, key=client-0
? Envoyé ? partition=2, offset=4, key=client-0
? Envoyé ? partition=2, offset=5, key=client-0
? Envoyé ? partition=1, offset=3, key=client-2
? Envoyé ? partition=1, offset=4, key=client-2
? Envoyé ? partition=1, offset=5, key=client-2
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.237 s
[INFO] Finished at: 2026-05-02T21:26:35+01:00
[INFO] -----
PS C:\tp-kafka-java>
```

```

? Envoyé ? partition=1, offset=297, key=client-2
? Envoyé ? partition=1, offset=298, key=client-2
? Envoyé ? partition=1, offset=299, key=client-2
? Envoyé ? partition=1, offset=300, key=client-2
? Envoyé ? partition=1, offset=301, key=client-2
? Envoyé ? partition=1, offset=302, key=client-2
? Envoyé ? partition=1, offset=303, key=client-2
? Envoyé ? partition=1, offset=304, key=client-2
? Envoyé ? partition=1, offset=305, key=client-2
? Envoyé ? partition=1, offset=306, key=client-2
? Envoyé ? partition=1, offset=307, key=client-2
? Envoyé ? partition=1, offset=308, key=client-2
? Envoyé ? partition=1, offset=309, key=client-2
? Envoyé ? partition=1, offset=310, key=client-2
? Envoyé ? partition=1, offset=311, key=client-2
? Envoyé ? partition=1, offset=312, key=client-2
? Envoyé ? partition=1, offset=313, key=client-2
? Envoyé ? partition=1, offset=314, key=client-2
? Envoyé ? partition=1, offset=315, key=client-2
? Envoyé ? partition=1, offset=316, key=client-2
? Envoyé ? partition=1, offset=317, key=client-2
? Envoyé ? partition=1, offset=318, key=client-2
? Envoyé ? partition=1, offset=319, key=client-2
? Envoyé ? partition=1, offset=320, key=client-2
? Envoyé ? partition=1, offset=321, key=client-2
? Envoyé ? partition=1, offset=322, key=client-2
? Envoyé ? partition=1, offset=323, key=client-2
? Envoyé ? partition=1, offset=324, key=client-2
? Envoyé ? partition=1, offset=325, key=client-2
? Envoyé ? partition=1, offset=326, key=client-2
? Envoyé ? partition=1, offset=327, key=client-2
? Envoyé ? partition=1, offset=328, key=client-2
? Envoyé ? partition=1, offset=329, key=client-2
? Envoyé ? partition=1, offset=330, key=client-2
? Envoyé ? partition=1, offset=331, key=client-2
? Envoyé ? partition=1, offset=332, key=client-2
? Envoyé ? partition=1, offset=333, key=client-2
? Envoyé ? partition=1, offset=334, key=client-2
? Envoyé ? partition=1, offset=335, key=client-2
? Envoyé ? partition=1, offset=336, key=client-2
? Envoyé ? partition=1, offset=337, key=client-2
? Envoyé ? partition=1, offset=338, key=client-2
[NUP] -----
[NUP] BUILD SUCCESS
[NUP] -----
[NUP] Total time: 3.872 s
[NUP] Finished at: 2026-05-02T21:28:09+01:00
[NUP] -----

```

Réponses Exercice 4.B

Question 1 — Sans idempotence, que se passe-t-il ?

Si Kafka s'arrête pendant l'envoi, le producer retry automatiquement (3 fois). Sans idempotence, un message peut être envoyé plusieurs fois → des doublons apparaissent dans le topic.

Question 2 — Avec idempotence, quelle différence ?

Réactive true dans SimpleProducer.java :

```
props.put(ProducerConfig.ENABLE_IDEMPOTENCE_CONFIG, true);
```

Kafka assigne un numéro de séquence unique à chaque message. Si le broker reçoit deux fois le même message, il ignore le doublon automatiquement.

Question 3 — Rôle des numéros de séquence (KIP-98) :

Chaque producer reçoit un Producer ID (PID). Chaque message a un numéro de séquence. Le broker vérifie : si séquence déjà reçue → message ignoré. Cela garantit exactly-once au niveau du producer.

```

--release 17 is recommended instead of -source 17 -target 17 because it sets the location of sys
cally
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ tp-kafka-java ---
[INFO] skip non existing resourceDirectory C:\tp-kafka-java\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ tp-kafka-java ---
[INFO] Recompiling the module because of changed dependency.
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ tp-kafka-java ---
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ tp-kafka-java ---
[INFO] Building jar: C:\tp-kafka-java\target\tp-kafka-java-1.0-SNAPSHOT.jar
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 2.418 s
[INFO] Finished at: 2026-05-02T21:30:18+01:00
[INFO] -----
PS C:\tp-kafka-java>

```

Exercise 4.C — S rialisation JSON

```

[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ tp-kafka-java ---
[INFO] skip non existing resourceDirectory C:\tp-kafka-java\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ tp-kafka-java ---
[INFO] Recompiling the module because of changed dependency.
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ tp-kafka-java ---
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ tp-kafka-java ---
[INFO] Building jar: C:\tp-kafka-java\target\tp-kafka-java-1.0-SNAPSHOT.jar
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 11.202 s
[INFO] Finished at: 2026-05-02T21:31:58+01:00
[INFO] -----
PS C:\tp-kafka-java> notepad src\main\java\tn\utm\kafka\Commande.java
PS C:\tp-kafka-java> notepad src\main\java\tn\utm\kafka\CommandeProducer.java
PS C:\tp-kafka-java>

```



```
PS C:\WINDOWS\system32> C:\kafka\bin\windows\kafka-topics.bat --bootstrap-server localhost:9092 --create --topic commandes --partitions 4 --replication-factor 1
Created topic commandes.
PS C:\WINDOWS\system32>
```

```
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
? En attente de commandes... (Ctrl+C pour arrêter)
=====
? ID      : CMD-002
? Date    : 2026-05-02
? Articles : [Article-A, Article-B]
? Total   : 200.0 DT
=====
? ID      : CMD-005
? Date    : 2026-05-02
? Articles : [Article-A, Article-B]
? Total   : 500.0 DT
=====
? ID      : CMD-001
? Date    : 2026-05-02
? Articles : [Article-A, Article-B]
? Total   : 100.0 DT
=====
? ID      : CMD-003
? Date    : 2026-05-02
? Articles : [Article-A, Article-B]
? Total   : 300.0 DT
=====
? ID      : CMD-004
? Date    : 2026-05-02
? Articles : [Article-A, Article-B]
? Total   : 400.0 DT
```

Partie 5 — Partitions, offsets, garanties de livraison

Exercice 5.A — Mesurer le LAG sous charge

```
PS C:\tp-kafka-java> C:\kafka\bin\windows\kafka-topics.bat --bootstrap-server localhost:9092 --create --topic stress-test --partitions 6 --replication-factor 1
Created topic stress-test.
PS C:\tp-kafka-java>
```

```
PS C:\WINDOWS\system32> C:\kafka\bin\windows\kafka-consumer-groups.bat --bootstrap-server localhost:9092 --describe --group perf-1
```

GROUP	TOPIC	PARTITION	CURRENT-OFFSET	LOG-END-OFFSET	LAG	CONSUMER-ID
perf-1	stress-test	0	107396	108373	977	console-consumer-0d693918-f2c
5-4b47-b6e8-589b7a0cbe22 /127.0.0.1			console-consumer			
perf-1	stress-test	2	105167	106573	1406	console-consumer-0d693918-f2c
5-4b47-b6e8-589b7a0cbe22 /127.0.0.1			console-consumer			
perf-1	stress-test	1	106874	107839	965	console-consumer-0d693918-f2c
5-4b47-b6e8-589b7a0cbe22 /127.0.0.1			console-consumer			
perf-1	stress-test	4	96184	97556	1372	console-consumer-0d693918-f2c
5-4b47-b6e8-589b7a0cbe22 /127.0.0.1			console-consumer			
perf-1	stress-test	3	104099	104758	659	console-consumer-0d693918-f2c
5-4b47-b6e8-589b7a0cbe22 /127.0.0.1			console-consumer			
perf-1	stress-test	5	99200	99502	302	console-consumer-0d693918-f2c
5-4b47-b6e8-589b7a0cbe22 /127.0.0.1			console-consumer			

```
PS C:\WINDOWS\system32>
```

Avec 1 seul consumer → LAG augmente car il lit toutes les 6 partitions seul. Avec 6 consumers → chaque consumer prend 1 partition → LAG diminue rapidement !

Le LAG augmente-t-il, diminue-t-il, ou reste-t-il stable ?

Configuration	Comportement du LAG
1 consumer	LAG augmente — 1 consumer lit 6 partitions seul, il n'arrive pas à suivre les 10 000 msg/sec
6 consumers	LAG diminue rapidement — 1 consumer par partition, traitement parallèle maximal

Exercice 5.B — Topic compacté

```
PS C:\tp-kafka-java> C:\kafka\bin\windows\kafka-topics.bat --bootstrap-server localhost:9092 --create --topic etat-utilisateurs --partitions 3 --replication-factor 1 --config cleanup.policy=compact
Created topic etat-utilisateurs.
PS C:\tp-kafka-java>
```

```
PS C:\WINDOWS\system32> C:\kafka\bin\windows\kafka-console-producer.bat --bootstrap-server localhost:9092 --topic etat-utilisateurs --property "parse.key=true" --property "key.separator=:"
>user-42:{"nom":"Heithem","pays":"TN"}
>user-42:{"nom":"Heithem","pays":"TN","ville":"Tunis"}
>user-42:{"nom":"Heithem","pays":"TN","ville":"Tunis","role":"admin"}
```



```

PS C:\WINDOWS\system32> C:\kafka\bin\windows\kafka-console-producer.bat --bootstrap-server localhost:9092 --topic et
tilisateurs --property "parse.key=true" --property "key.separator=:"
>user-42:
>Terminer le programme de commandes (O/N) ?
Terminer le programme de commandes (O/N) ? o
PS C:\WINDOWS\system32> C:\kafka\bin\windows\kafka-console-consumer.bat --bootstrap-server localhost:9092 --topic et
tilisateurs --from-beginning --property "print.key=true" --property "key.separator=:"
user-42:{"nom":"Heithem","pays":"TN"}
user-42:{"nom":"Heithem","pays":"TN","ville":"Tunis"}
user-42:{"nom":"Heithem","pays":"TN","ville":"Tunis","role":"admin"}
user-42:

```

```

PS C:\WINDOWS\system32> taskkill /F /PID 17600
Opération réussie : le processus avec PID 17600 a été terminé.
PS C:\WINDOWS\system32> C:\kafka\bin\windows\kafka-consumer-groups.bat --bootstrap-server localhost:9092 --describe -
oup ca-1

```

GROUP	TOPIC	HOST	PARTITION	CURRENT-OFFSET	LOG-END-OFFSET	LAG	CONSUMER-ID
ca-1	pos-events	0	3478	3478	0	consumer-ca-1-1-314a2ac2-5	
-4431-8ca9-2ebf26082e26	/127.0.0.1	consumer-ca-1-1					
ca-1	pos-events	1	3607	3607	0	consumer-ca-1-1-314a2ac2-5	
-4431-8ca9-2ebf26082e26	/127.0.0.1	consumer-ca-1-1					
ca-1	pos-events	3	0	0	0	consumer-ca-1-1-8721cf20-b	
-4c65-85ff-5c580eefd957	/127.0.0.1	consumer-ca-1-1					
ca-1	pos-events	2	1752	1752	0	consumer-ca-1-1-5352f073-9	
4a75-8a9c-da4570b7092a	/127.0.0.1	consumer-ca-1-1					

À quel moment exactement la compaction se déclenche-t-elle ?

La compaction ne se déclenche **pas immédiatement**. Elle se déclenche quand :

- Le ratio de messages "sales" dépasse log.cleaner.min.cleanable.ratio (défaut = **0.5** soit 50%)
- Le log cleaner thread tourne en arrière-plan toutes les **15 secondes** environ
- Après compaction : seule la **dernière valeur** de chaque clé est conservée
- Après le tombstone : la clé user-42 **disparaît complètement**

Partie 6 — Mini-projet : pipeline de logs en temps réel

```
package tn.utm.kafka;
```

```
import com.fasterxml.jackson.databind.JsonNode;
import com.fasterxml.jackson.databind.ObjectMapper;
import org.apache.kafka.clients.consumer.*;
import org.apache.kafka.common.serialization.StringDeserializer;
```

```
import java.time.Duration;
import java.util.*;
```

```
public class ChiffreAffairesParVille {
```

```
    private static final String TOPIC          = "pos-events";
    private static final String GROUP_ID       = "ca-1";
    private static final String BOOTSTRAP_SERVERS = "localhost:9092";
```

```
    private static final Map<String, Double> caParVille = new HashMap<>();
    private static final ObjectMapper mapper = new ObjectMapper();
    private static long dernierAffichage = System.currentTimeMillis();
```

```
    public static void main(String[] args) {
```

```
        Properties props = new Properties();
        props.put(ConsumerConfig.BOOTSTRAP_SERVERS_CONFIG, BOOTSTRAP_SERVERS);
        props.put(ConsumerConfig.GROUP_ID_CONFIG, GROUP_ID);
        props.put(ConsumerConfig.KEY_DESERIALIZER_CLASS_CONFIG, StringDeserializer.class.getName());
        props.put(ConsumerConfig.VALUE_DESERIALIZER_CLASS_CONFIG, StringDeserializer.class.getName());
```

<

```
package tn.utm.kafka;
```

```
import com.fasterxml.jackson.databind.JsonNode;
import com.fasterxml.jackson.databind.ObjectMapper;
import org.apache.kafka.clients.consumer.*;
import org.apache.kafka.clients.producer.*;
import org.apache.kafka.common.serialization.StringDeserializer;
import org.apache.kafka.common.serialization.StringSerializer;
```

```
import java.time.Duration;
import java.util.*;
```

```
public class DetecteurAnomalies {
```

```
    private static final String TOPIC_SOURCE    = "pos-events";
    private static final String TOPIC_ALERTES   = "alertes-retours";
    private static final String GROUP_ID        = "alerte-1";
    private static final String BOOTSTRAP_SERVERS = "localhost:9092";
```

```
    private static final ObjectMapper mapper = new ObjectMapper();
```

```
    public static void main(String[] args) {
```

```
        // === CONSUMER ===
        Properties consumerProps = new Properties();
        consumerProps.put(ConsumerConfig.BOOTSTRAP_SERVERS_CONFIG, BOOTSTRAP_SERVERS);
        consumerProps.put(ConsumerConfig.GROUP_ID_CONFIG, GROUP_ID);
```

```
package tn.utm.kafka;

import org.apache.kafka.clients.producer.*;
import org.apache.kafka.common.serialization.StringSerializer;

import java.util.Properties;
import java.util.Random;

public class SimulateurCaisse {

    private static final String TOPIC = "pos-events";
    private static final String BOOTSTRAP_SERVERS = "localhost:9092";

    private static final String[] VILLES = {"Tunis", "Sousse", "Sfax", "Bizerte", "Gabès"};
    private static final Random random = new Random();

    public static void main(String[] args) throws InterruptedException {

        Properties props = new Properties();
        props.put(ProducerConfig.BOOTSTRAP_SERVERS_CONFIG, BOOTSTRAP_SERVERS);
        props.put(ProducerConfig.KEY_SERIALIZER_CLASS_CONFIG, StringSerializer.class.getName());
        props.put(ProducerConfig.VALUE_SERIALIZER_CLASS_CONFIG, StringSerializer.class.getName());
        props.put(ProducerConfig.ENABLE_IDEMPOTENCE_CONFIG, "true");
        props.put(ProducerConfig.ACKS_CONFIG, "all");

        KafkaProducer<String, String> producer = new KafkaProducer<>(props);
```

```

Envoyé -> {"type":"VENTE","idCaisse":"CAISSE-SFAX-3","ville":"Sfax","timestamp":"2026-05-02T21:11:59.596549100Z","montant":159.05507842813873} | partition=0 | offset=1
Envoyé -> {"type":"VENTE","idCaisse":"CAISSE-BIZERTE-6","ville":"Bizerte","timestamp":"2026-05-02T21:11:59.795359700Z","montant":100.20192150258248} | partition=1 | offset=1
Envoyé -> {"type":"OUVERTURE","idCaisse":"CAISSE-GABÈS-7","ville":"Gabès","timestamp":"2026-05-02T21:12:00.141152400Z","montant":0.0} | partition=2 | offset=2
Envoyé -> {"type":"OUVERTURE","idCaisse":"CAISSE-GABÈS-7","ville":"Gabès","timestamp":"2026-05-02T21:12:00.309940200Z","montant":0.0} | partition=2 | offset=3
Envoyé -> {"type":"VENTE","idCaisse":"CAISSE-TUNIS-10","ville":"Tunis","timestamp":"2026-05-02T21:12:00.434993Z","montant":432.5473117681857} | partition=1 | offset=2
Envoyé -> {"type":"VENTE","idCaisse":"CAISSE-SFAX-2","ville":"Sfax","timestamp":"2026-05-02T21:12:00.619141400Z","montant":288.6553125870261} | partition=0 | offset=2
Envoyé -> {"type":"VENTE","idCaisse":"CAISSE-GABÈS-6","ville":"Gabès","timestamp":"2026-05-02T21:12:01.104911600Z","montant":273.88488150496534} | partition=2 | offset=4
Envoyé -> {"type":"RETOUR","idCaisse":"CAISSE-SFAX-9","ville":"Sfax","timestamp":"2026-05-02T21:12:01.329912Z","montant":308.33587285634513} | partition=0 | offset=3
Envoyé -> {"type":"RETOUR","idCaisse":"CAISSE-SFAX-5","ville":"Sfax","timestamp":"2026-05-02T21:12:01.592911900Z","montant":387.3108414033666} | partition=0 | offset=4
Envoyé -> {"type":"VENTE","idCaisse":"CAISSE-TUNIS-9","ville":"Tunis","timestamp":"2026-05-02T21:12:01.758687900Z","montant":95.98647377576363} | partition=1 | offset=3
Envoyé -> {"type":"VENTE","idCaisse":"CAISSE-BIZERTE-9","ville":"Bizerte","timestamp":"2026-05-02T21:12:02.115858500Z","montant":330.1874316607869} | partition=1 | offset=4
Envoyé -> {"type":"VENTE","idCaisse":"CAISSE-GABÈS-1","ville":"Gabès","timestamp":"2026-05-02T21:12:02.227874900Z","montant":263.4991991905458} | partition=2 | offset=5
Envoyé -> {"type":"VENTE","idCaisse":"CAISSE-SOUSSE-9","ville":"Sousse","timestamp":"2026-05-02T21:12:02.573875100Z","montant":64.10785200739406} | partition=0 | offset=5
Envoyé -> {"type":"VENTE","idCaisse":"CAISSE-TUNIS-7","ville":"Tunis","timestamp":"2026-05-02T21:12:02.760666100Z","montant":456.0294101486684} | partition=1 | offset=5
Envoyé -> {"type":"VENTE","idCaisse":"CAISSE-BIZERTE-1","ville":"Bizerte","timestamp":"2026-05-02T21:12:03.034393700Z","montant":338.8576153352124} | partition=1 | offset=6
Envoyé -> {"type":"RETOUR","idCaisse":"CAISSE-GABÈS-7","ville":"Gabès","timestamp":"2026-05-02T21:12:03.511998100Z","montant":246.0867462095866} | partition=2 | offset=6
Envoyé -> {"type":"VENTE","idCaisse":"CAISSE-SOUSSE-6","ville":"Sousse","timestamp":"2026-05-02T21:12:03.847868300Z","montant":188.32314370215806} | partition=0 | offset=6
Envoyé -> {"type":"RETOUR","idCaisse":"CAISSE-SFAX-8","ville":"Sfax","timestamp":"2026-05-02T21:12:04.026604300Z","montant":443.3533558429466} | partition=0 | offset=7
Envoyé -> {"type":"VENTE","idCaisse":"CAISSE-GABÈS-2","ville":"Gabès","timestamp":"2026-05-02T21:12:04.275339900Z","montant":125.21482011696193} | partition=2 | offset=7
Envoyé -> {"type":"RETOUR","idCaisse":"CAISSE-BIZERTE-8","ville":"Bizerte","timestamp":"2026-05-02T21:12:04.485065500Z","montant":85.87613633648905} | partition=1 | offset=7
Envoyé -> {"type":"VENTE","idCaisse":"CAISSE-SFAX-1","ville":"Sfax","timestamp":"2026-05-02T21:12:04.820058Z","montant":488.6743365853925} | partition=0 | offset=8
Envoyé -> {"type":"VENTE","idCaisse":"CAISSE-SOUSSE-2","ville":"Sousse","timestamp":"2026-05-02T21:12:05.195059100Z","montant":125.21482011696193} | partition=2 | offset=7

```

```

PS C:\WINDOWS\system32> C:\kafka\bin\windows\kafka-consumer-groups.bat --bootstrap-server localhost:9092 --describe --group ca-1
>> C:\kafka\bin\windows\kafka-consumer-groups.bat --bootstrap-server localhost:9092 --describe --group alerte-1

```

GROUP	TOPIC	HOST	PARTITION	CURRENT-OFFSET	LOG-END-OFFSET	LAG	CONSUMER-ID
ca-1	pos-events	0	814	814	0	consumer-ca-1-1-314a2ac2-55e2	
-4431-8ca9-2ebf26082e26	/127.0.0.1	consumer-ca-1-1					
ca-1	pos-events	1	849	849	0	consumer-ca-1-1-314a2ac2-55e2	
-4431-8ca9-2ebf26082e26	/127.0.0.1	consumer-ca-1-1					
ca-1	pos-events	2	426	426	0	consumer-ca-1-1-5352f073-9393	
-4a75-8afc-da4570b7092a	/127.0.0.1	consumer-ca-1-1					
ca-1	pos-events	3	0	0	0	consumer-ca-1-1-5352f073-9393	
-4a75-8afc-da4570b7092a	/127.0.0.1	consumer-ca-1-1					
GROUP	TOPIC	HOST	PARTITION	CURRENT-OFFSET	LOG-END-OFFSET	LAG	CONSUMER-ID
alerte-1	pos-events	2	429	429	0	consumer-alerte-1-1-b122442d-	
c669-462e-b2cd-86e7bb2caff7	/127.0.0.1	consumer-alerte-1-1					
alerte-1	pos-events	3	0	0	0	consumer-alerte-1-1-b122442d-	
c669-462e-b2cd-86e7bb2caff7	/127.0.0.1	consumer-alerte-1-1					
alerte-1	pos-events	0	820	820	0	consumer-alerte-1-1-b122442d-	
c669-462e-b2cd-86e7bb2caff7	/127.0.0.1	consumer-alerte-1-1					
alerte-1	pos-events	1	854	854	0	consumer-alerte-1-1-b122442d-	
c669-462e-b2cd-86e7bb2caff7	/127.0.0.1	consumer-alerte-1-1					

```

PS C:\WINDOWS\system32>

```

```
PS C:\WINDOWS\system32> C:\kafka\bin\windows\kafka-consumer-groups.bat --bootstrap-server localhost:9092 --describe --group ca-1
```

GROUP	TOPIC	HOST	PARTITION	CURRENT-OFFSET	LOG-END-OFFSET	LAG	CONSUMER-ID
ca-1	pos-events	0	1237	1237	0	consumer-ca-1-1-314a2ac2-55e2	
-4431-8ca9-2ebf26082e26 /127.0.0.1			consumer-ca-1-1				
ca-1	pos-events	1	1293	1293	0	consumer-ca-1-1-314a2ac2-55e2	
-4431-8ca9-2ebf26082e26 /127.0.0.1			consumer-ca-1-1				
ca-1	pos-events	3	0	0	0	consumer-ca-1-1-8721cf20-b249	
-4c65-85ff-5c580eefd957 /127.0.0.1			consumer-ca-1-1				
ca-1	pos-events	2	643	643	0	consumer-ca-1-1-5352f073-9393	
-4a75-8afc-da4570b7092a /127.0.0.1			consumer-ca-1-1				

```
PS C:\WINDOWS\system32>
```

2. ANALYSE — REBALANCE

◆ Analyse du comportement de Kafka lors des rebalances

Le mécanisme de **rebalance** dans Kafka se produit lorsqu'un changement intervient dans un groupe de consommateurs, notamment dans les cas suivants :

- Ajout d'un nouveau consumer
- Arrêt ou crash d'un consumer
- Modification du nombre de partitions

Lors d'un rebalance, Kafka redistribue les partitions du topic entre les consommateurs actifs du groupe afin d'assurer une répartition équilibrée de la charge.

◆ Impact du rebalance

Pendant le rebalance :

- La consommation est temporairement interrompue
- Les partitions sont réassignées
- Les offsets sont réinitialisés ou récupérés

Cela peut entraîner :

- Une latence temporaire
- Une duplication de traitement (dans certains cas)
- Une pause dans le flux de données

⚠ 3. LIMITES DU MONO-BROKER

◆ Limites du déploiement mono-broker

Dans ce TP, Kafka est déployé avec un seul broker, ce qui présente plusieurs limitations importantes :

1. Absence de tolérance aux pannes

- Si le broker tombe, tout le système est indisponible
- Aucun mécanisme de réplication n'est possible

2. Pas de haute disponibilité

- Un seul point de défaillance (Single Point of Failure)
- Impossible d'assurer la continuité de service

3. Scalabilité limitée

- Les performances sont limitées aux ressources d'une seule machine
- Impossible de répartir la charge sur plusieurs serveurs

4. Réplication impossible

- Le facteur de réplication est forcé à 1
- Risque de perte de données élevé

5. Tests non réalistes

- Les conditions de production (multi-brokers) ne sont pas reproduites
- Certaines fonctionnalités avancées (ISR, failover) ne peuvent pas être testées